

Lower Bounding Techniques for DSATUR-based Branch and Bound

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Abstract

Given an undirected graph, the Vertex Coloring Problem (VCP) consists of assigning a color to each vertex of the graph such that two adjacent vertices do not share the same color and the total number of colors is minimized. DSATUR-based Branch-and-Bound is a well-known exact algorithm for the VCP. One of its main drawbacks is that a lower bound (equal to the size of a maximal clique) is computed once at the root of the branching scheme and it is never updated during the execution of the algorithm. In this article, we show how to update the lower bound and we compare the efficiency of several lower bounding techniques.

Keywords: Graph Coloring, DSATUR, Branch and Bound.

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1 Introduction

The *Vertex Coloring Problem* (VCP) is one of the basic problems in graph theory with application in many areas, including: scheduling, timetabling, register allocation, frequency assignment, communication networks and many others. Given an undirected graph $G = (V, E)$ with $|V|$ vertices and $|E|$ edges, a *stable set* of G is a subset of non-connected vertices and a *clique* of G is a subset of fully connected vertices. The *stability number* $\alpha(G)$ is the size of the maximum stable set in G . The *clique number* $\omega(G)$ is the size of the maximum clique in G . A *coloring* C of G is a partition of V into k non empty stable sets: $C = \{V_1, \dots, V_k\}$. All vertices belonging to V_i are colored with the same color i . The *chromatic number* of G , denoted by $\chi(G)$, is the minimum number of stable sets (or equivalently colors) in a coloring of G . The VCP is the problem of determining the chromatic number of G . A *partial coloring* of G is a partition of $\tilde{V} \subset V$ into $\tilde{k}$ stable sets (i.e., the number of colors): $\tilde{C} = \{\tilde{V}_1, \dots, \tilde{V}_{\tilde{k}}\}$. The remaining vertices $V \setminus \tilde{V}$ are non-colored. Given a partial coloring $\tilde{C}$ of G , we denote by $\chi_{\tilde{C}}(G)$ the chromatic number of G partially colored by the partial coloring $\tilde{C}$.

The VCP has received a large amount of attention in the last decades. A complete literature review goes beyond the scope of this paper. We refer the reader to Malaguti and Toth [5] for a complete survey on the topic. It is worth mentioning that exact algorithms based on Column Generation and Branch-and-Price (B&P) are particularly effective for “hard” VCP instances, e.g. [3], [4], [6], [7] and [8]. For random graphs instead DSATUR-based Branch and Bound has proven to outperform B&P see [10]. In this paper we study several lower bounding techniques to improve the performances of DSATUR-based Branch and Bound (see [1,10,12]). A second contribution of this paper is to compute and compare the values of a set of lower bounds for the VCP: the clique number, the Lovász Theta number and an upper bound on the stability number of an auxiliary graph (see Section 3). Many theoretical studies have been done on this topic but, to the best of our knowledge, none have experimentally compared the values of those bounds.

2 DSATUR-based Branch and Bound

In this section, we describe the exact implicit enumeration algorithm based on DSATUR, called *DSATUR-based Branch and Bound* (DSATUR-B&B). DSATUR (see Brélaz [1]) is a greedy heuristic algorithm where each vertex $u \in V$ is iteratively colored with a feasible color. Given a (partial) coloring $\tilde{C}$,

the *saturation degree* $\text{DSAT}(u)$ of a vertex $u \in V$ corresponds to the number of different colors in its neighbourhood $N(u)$. At each iteration of DSATUR, the vertex with the highest DSAT value is colored. An exact branch and bound algorithm can be derived from DSATUR as follows. At each node of the branch and bound tree, we are given a (partial) coloring $\tilde{C}$ with $\tilde{k}$ colors, an upper bound (UB) and a lower bound (LB) on the chromatic number. After a non-colored vertex has been selected, up to $\tilde{k} + 1$ children nodes are created by coloring the vertex with all the feasible colors in $\tilde{C}$ plus a new one.

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DSATUR-B&B( $\tilde{C}$ )
If  $LB \geq UB \rightarrow$  Return;
If all the vertices are colored Then
    If  $\tilde{k} < UB$  Then  $UB = \tilde{k}$ 
Else
    select a non-colored vertex  $v$ 
    For every feasible color  $i \in \tilde{C}$  plus a new one,
         $\hat{C} \leftarrow \tilde{C}$ , add  $v$  in  $\hat{V}_i$ ;
        DSATUR-B&B( $\hat{C}$ )
    End For
End If

```

Many vertex selection rules have been introduced in the literature (see [1,12]). The most effective one has been proposed in [10] and it allows a sensitive reduction of the branching nodes. In all the implementations (see [1,12,10]), the lower bound LB is initialized at the beginning of the algorithm by using the cardinality of a maximal clique in G . This LB is then never updated during the execution of the algorithm. While most of the works in the literature tries to improve DSATUR-B&B focusing on the vertex selection rule, our idea instead is to improve the lower bound, updating its value during the branching scheme.

3 Lower bounds for the Vertex Coloring Problem

In the following we present the three lower bounds for the VCP considered in this paper, i.e., the clique number, the Lovász Theta number and an upper bound on the stability number of an auxiliary graph. In order to make these bounds dependent on a partial coloring $\tilde{C}$ obtained during the execution of DSATUR-B&B, we compute them on a smaller graph. This *Reduced Graph* $G^{\tilde{C}} = (V^{\tilde{C}}, E^{\tilde{C}})$ is composed of the sub-graph of G induced by the non-colored

vertices plus $\tilde{k}$ vertices, one for each color. Each new vertex $\tilde{v}_i$, representing color i , is connected to all the non-colored neighbours of the vertices of V_i and to all the others $\tilde{k} - 1$ new vertices. Thus, the subgraph of $G^{\tilde{C}}$ induced by the $\tilde{k}$ new vertices is a clique. The reduced graph becomes smaller increasing the number of colored vertices thus also the lower bounds become easier to compute. An example using a partially colored graph of 6 vertices is given in Figure 1, where two colors (1 and 2) are used and three vertices are non-colored. The Reduced Graph has 5 vertices, two representing the classes of colors plus the three original non-colored vertices.

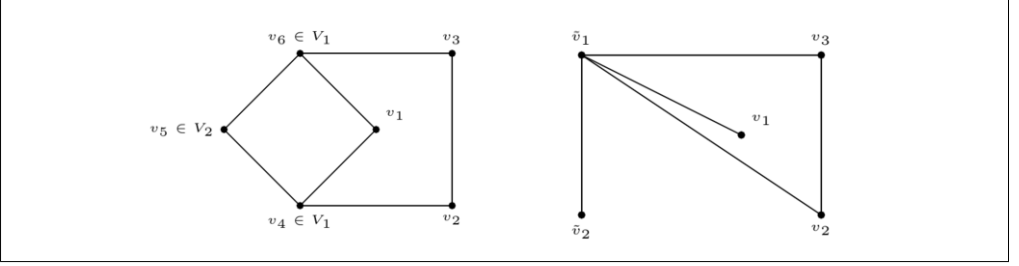


Fig. 1: A partially colored graph G and the Reduced Graph $G^{\tilde{C}}$

The following holds:

Lemma 3.1 $\chi_{\tilde{C}}(G) = \chi_{\tilde{C}}(G^{\tilde{C}})$

Proof. Any feasible coloring $\hat{C} = \{\hat{C}_1, \dots, \hat{C}_k\}$ in G containing the coloring $\tilde{C}$ (inducing $k \geq \tilde{k}$) can be mapped into an unique feasible coloring $\{\hat{C}'_1, \dots, \hat{C}'_k\}$ in $G^{\tilde{C}}$ using the same number of colors k . For each color i , $i = 1, \dots, k$, two cases arise: if $\hat{C}_i \subseteq V \setminus \tilde{C}$, then $\hat{C}'_i \leftarrow \hat{C}_i$ (since the subgraphs induced by $\hat{C}_i$ are equivalent in G and $G^{\tilde{C}}$), otherwise it exists a unique stable set j such that $\tilde{C}_j \subseteq \hat{C}_i$, then $\hat{C}'_i \leftarrow \{\tilde{v}_j\} \cup \{\hat{C}_i \setminus \tilde{C}_j\}$ with $\tilde{v}_j$ representing $\tilde{C}_j$ in $G^{\tilde{C}}$. By construction, $\hat{C}'_i$ is a stable set $G^{\tilde{C}}$. Any feasible coloring $\{\hat{C}'_1, \dots, \hat{C}'_k\}$ in $G^{\tilde{C}}$ can be mapped into an unique feasible coloring $\{\hat{C}_1, \dots, \hat{C}_k\}$ containing the coloring $\tilde{C}$ in G . For each color i , $i = 1, \dots, k$, two cases arise: if $\hat{C}'_i \subseteq V \setminus \tilde{C}$, then $\hat{C}_i \leftarrow \hat{C}'_i$, otherwise an unique vertex $\tilde{v}_j$ representing $\tilde{C}_j$ belongs to $\hat{C}'_i$ (let us recall that the $\tilde{k}$ vertices representing $\tilde{C}$ in $G^{\tilde{C}}$ form a clique), and then $\hat{C}_i \leftarrow \tilde{C}_j \cup \{\hat{C}'_i \setminus \{\tilde{v}_j\}\}$. By construction, $\hat{C}_i$ is a stable set in G . $\square$

Trivially, we have: $\chi_{\tilde{C}}(G^{\tilde{C}}) = \chi(G^{\tilde{C}})$. Thus, from lemma 3, a lower bound denoted by $LB^{\tilde{C}}$ for $\chi(G^{\tilde{C}})$ is a lower bound for $\chi_{\tilde{C}}(G)$. Given a partial coloring $\tilde{C}$, the following values are valid lower bounds for $\chi(G^{\tilde{C}})$:

Clique number. A first lower bound is the clique number $\omega(G^{\tilde{C}})$ of the Reduced Graph. Computing this bound is NP-hard but in practice very effective exact methods are available in the literature.

Lovász Theta number. The Lovász Theta number, denoted $\vartheta(\tilde{G}^{\tilde{C}})$ ($n = |V^{\tilde{C}}|$), on the complementary graph $\tilde{G}^{\tilde{C}}$ is a second lower bound. This bound can be computed by solving the following positive semi-definite program (SDP):

$$\vartheta(\tilde{G}^{\tilde{C}}) = \max \sum_{i=1}^n Y_{0i} : Y_{0i} = Y_{ii}, Y_{ij} = 0, (ij) \in E^{\tilde{C}}, Y \in S_+^{n+1} \quad (1)$$

where Y_{ij} ($i, j \in V^{\tilde{C}}$) represents the product of two binary variables associated to the vertices and S_+^{n+1} denotes the cone of real symmetric square positive semidefinite matrices of order $n+1$. This SDP is solvable in polynomial time and many SDP effective solvers are present in the literature.

Stability number of an auxiliary graph. A third lower bound can be obtained using an auxiliary graph introduced in Cornaz and Jost (2008) [2]. This auxiliary graph, denoted $G_A^{\tilde{C}}$, allows a 1-to-1 mapping between stable sets in $G_A^{\tilde{C}}$ and colorings in $G^{\tilde{C}}$. The following equality holds: $\chi(G^{\tilde{C}}) = |V^{\tilde{C}}| - \alpha(G_A^{\tilde{C}})$. The auxiliary graph is defined into 3 steps : first step, define an acyclic orientation of the complementary graph $\tilde{G}^{\tilde{C}}$; second step, define the line-graph of this acyclic digraph; third step, remove all edges between pairs of arcs which are simplicial in the line-graph (see [2] for more details). Any upper bound of the stability number $\bar{\alpha}(G_A^{\tilde{C}})$ defines a valid lower bound since: $\chi(G^{\tilde{C}}) \geq |V| - \bar{\alpha}(G_A^{\tilde{C}})$.

4 Computational results

All algorithms are coded in C/C++, and run on a PC with an Intel(R) Core(TM) i7-4770 CPU at 3.40GHz and 16 GB RAM memory, under Linux Ubuntu 14.04 64-bit. The optimization software used in our test is Cplex 12.6 single thread and the SDP solver used is CSDP. The clique number is computed using Cliquer (see [9] for further details) and available on internet. To test our algorithm we used 10 instances introduced in [10] with 70 vertices and density varying from 0.1 to 0.9. On these set of instances, DSATUR-B&B has been shown to be the best exact algorithm (see [10]). In Table 1, for each instance we report the dimension $|V|$ and its density d . Then for the original graph G we report the chromatic number and the number of nodes for two

versions of DSATUR-B&B: the first one setting $UB = \infty$ and the second one setting the UB to the optimal value. We used the vertex selection rule proposed in [10]. Then the table reports the 3 lower bounds: the clique number ω , the Lovász Theta number ϑ and the bound $\bar{\alpha}$. The latter is computed using the value of the root node provided by CPLEX when solving the stable set problem on G_A , i.e. the linear programming relaxation of the stable set formulation with edge constraints and the additional cuts generated by default. For each bound we report the corresponding computing time.

In Table 1, we measure the relative computational strength of the bounds. The clique number is the fastest bound to compute despite the NP-hardness of the problem but also the less effective one. As expected, the Lovász Theta number dominates the clique number but the computational effort is considerably higher. Surprisingly the bound $\bar{\alpha}$ is always the strongest one and its computational time decreases as the density of the instances increases. This latter fact is due to the reduced dimension of the auxiliary graph when the density of the instances increases. As shown in Table 1, the number of nodes explored by DSATUR-B&B increases exponentially with the density of the instances. The use of lower bounding techniques is then crucial in order to speed up the computational convergence of the algorithm.

Instance			DSATUR-B&B									
$ V $	d	$\chi(G)$	Nodes	Time	Nodes*	Time*	$\omega(G)$	Time	$\vartheta(\bar{G})$	Time	$ V - \bar{\alpha}(G_A)$	Time
70	0.1	4	493	0.00	5	0.00	4	0.00	4	9.45	4	3.10
70	0.2	6	849	0.00	737	0.00	4	0.00	5	6.52	5	7.98
70	0.3	8	86866	0.23	86437	0.23	6	0.00	6	4.71	7	7.99
70	0.4	10	2700957	9.63	838240	2.98	7	0.00	8	2.81	9	5.63
70	0.5	12	3093345	14.05	3042810	13.45	9	0.00	10	2.17	11	3.27
70	0.6	14	2048941	12.31	1493746	8.66	10	0.00	12	1.17	13	1.66
70	0.7	17	5386833	40.90	447507	3.38	14	0.00	15	0.55	17	0.38
70	0.8	21	7140982	68.65	555393	5.20	17	0.00	19	0.24	21	0.10
70	0.9	28	2047390	23.12	2032323	22.56	24	0.00	26	0.10	28	0.01

Table 1
Different bounds for the original graph G

In Table 2, we measure the impact of the three lower bounding techniques based on updating the corresponding lower bound at each node of DSATUR-B&B. In our experiments the chromatic number is given in input. This allows to measure the impact of the LB while removing the impact of the UB. For each experiment, we set a time limit of 600 seconds (in case of time limit we report “-”). For each bound we report the absolute difference between the time without and with the use of the bounds (positive values means a speed-

up), the percentage of pruned nodes (a value of 100% means that the bound is able to prove optimality at the root node), the percentage of time required to compute the bounds over the total time.

As far as the clique number is concerned, the bound is able to cut a minimum of 30% and a maximum of 90% of nodes. However computing this bound at each node is computationally expensive and the overall computing time increases. The Lovász Theta number is computationally too expensive with the current status of the SDP solvers: the time limit is reached in almost all the instances. As far as the $\bar{\alpha}(G_A)$ bound is concerned, its value is able to prove optimality at the root node for high density instances. This is particularly promising since DSATUR-B&B tends to struggle when the density of the instance is high while the $\bar{\alpha}(G_A)$ bound becomes easier to compute. High density instances are difficult for DSATUR-B&B since the chromatic number is high while easy for the $\bar{\alpha}(G_A)$ since the auxiliary graph is small.

Instance		$\omega(G^{\tilde{C}})$			$\vartheta(\tilde{G}^{\tilde{C}})$			$ V^{\tilde{C}} - \bar{\alpha}(G_A^{\tilde{C}})$		
$ V $	d	diff time	% Nodes	% time	diff time	% Nodes	% time	diff time	% Nodes	% time
70	0.1	0.00	80.00	80	-9.22	80.00	100	-3.17	80.00	100
70	0.2	-0.04	30.39	91	-343.48	87.38	100	-330.65	90.37	100
70	0.3	-5.88	39.43	87	-	-	-	-	-	-
70	0.4	-78.14	53.28	83	-	-	-	-	-	-
70	0.5	-287.04	58.34	83	-	-	-	-	-	-
70	0.6	-141.91	65.43	84	-	-	-	-235.05	99.98	100
70	0.7	-24.53	82.18	86	-	-	-	2.99	100.00	100
70	0.8	-19.06	89.77	88	-	-	-	5.09	100.00	100
70	0.9	-90.11	87.06	86	-	-	-	23.11	100.00	98

Table 2
Effects of the lower bounding techniques

5 Conclusion

In this paper we have considered a series of different lower bounding techniques for DSATUR-B&B. The principal idea is to exploit lower bounds for the Vertex Coloring Problem in order to prune the implicit enumeration scheme. In the literature, all the efforts have been made in the direction of a better selection of the branching node. In this paper we have shown instead the potential of exploiting fast but strong lower bounds within the branching scheme. These bounds can potentially lead to a significant reduction of time and nodes for DSATUR-B&B. Future developments can be to develop more sophisticated

rules to trigger the bound computation only in the most promising part of the enumeration tree or when the gap between the lower and upper bound is low.

References

- [1] D. Brélaz, New methods to color the vertices of a graph. *Communications of the ACM (CACM)* 22(4): 251-256, 1979.
- [2] D. Cornaz, V. Jost, A one-to-one correspondence between colorings and stable sets. *Operations Research Letters* 36(6): 673-676, 2008.
- [3] S. Gualandi and F. Malucelli. Exact solution of graph coloring problems via constraint programming and column generation. *INFORMS Journal on Computing*, 24(1):81–100, 2012.
- [4] S. Held, W. Cook, and E. Sewell. Maximum-weight stable sets and safe lower bounds for graph coloring. *Mathematical Programming Computation*, 4(4):363–381, 2012.
- [5] E. Malaguti and P. Toth. A survey on vertex coloring problems. *International Transactions in Operational Research*, 17(1):1–34, 2010.
- [6] E. Malaguti, M. Monaci, and P. Toth. An exact approach for the vertex coloring problem. *Discrete Optimization*, 8(2):174 – 190, 2011.
- [7] A. Mehrotra, M. Trick, A column generation approach for graph coloring, *INFORMS Journal on Computing* 8(4): 344-54, 1996.
- [8] D. Morrison, J. Sauppe, E. Sewell, and S. Jacobson. A wide branching strategy for the graph coloring problem. *INFORMS Journal on Computing*, 26(4):704–717, 2014.
- [9] PRJ Ostergard. A fast algorithm for the maximum clique problem *Discrete Applied Mathematics*, 120(1):197–207, 2002.
- [10] P. San Segundo, A new DSATUR-based algorithm for exact vertex coloring. *Computers & Operations Research* 39(7): 1724-1733, 2012.
- [11] A. Schrijver, Combinatorial optimization, *Springer-Verlag, Berlin Heidelberg* : 1095-1096, 2002.
- [12] E. Sewell, An improved algorithm for exact graph coloring. *Cliques, coloring, and satisfiability. Proceedings of the second DIMACS implementation challenge*26: 359-373, 1996.